

DRWN TECHNOLOGY

ITEP
414

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Nature of Business

01

A software development company that provides web, mobile, and business software solutions for organizations.

Company Vision

02

To be a trusted technology company that delivers innovative and reliable digital solutions.

Company Mission

03

To develop high-quality software, provide excellent customer service, and promote continuous innovation through technology.

Office Location

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3rd Floor, DRWN Innovation Center
Tech Avenue, Santa Rosa City, Laguna, Philippines

Org Structure

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- Information Technology Department (5 Employees)
 - Human Resources Department (4 Employees)
 - Finance Department (5 Employees)
 - Sales Department (6 Employees)

Employee Distribution

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Department	Number of Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
Total	20

Part 2

Enterprise Hardware Inventory

Asset ID	Hardware	Quantity	Department	Purpose
DRWN-HW-001	Desktop Computer	15	HR, Finance, Sales	Daily business operations, document processing, and communication
DRWN-HW-002	Laptop	5	IT	System administration, remote troubleshooting, and software development
DRWN-HW-003	Enterprise Server	1	IT	Files sharing centralized management, and internal services
DRWN-HW-004	Business Router	1	IT	Internet access and network traffic routing
DRWN-HW-005	Switch	1	IT	Connects wired devices in the workplace
DRWN-HW-006	Printer	2	Shared Resource	Printing, Scanning and document management
DRWN-HW-007	UPS	2	IT	Power Protection
DRWN-HW-008	Wireless AP	3	Office-wide	Wireless connectivity
DRWN-HW-009	NAS Storage	1	IT	Centralized Storage
DRWN-HW-010	External Backup Drive	2	IT	Secondary Backup and recovery

DRWN-HW-011	Monitors	20	All Departments	Improving work qualtiy
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Enterprise Software Inventory

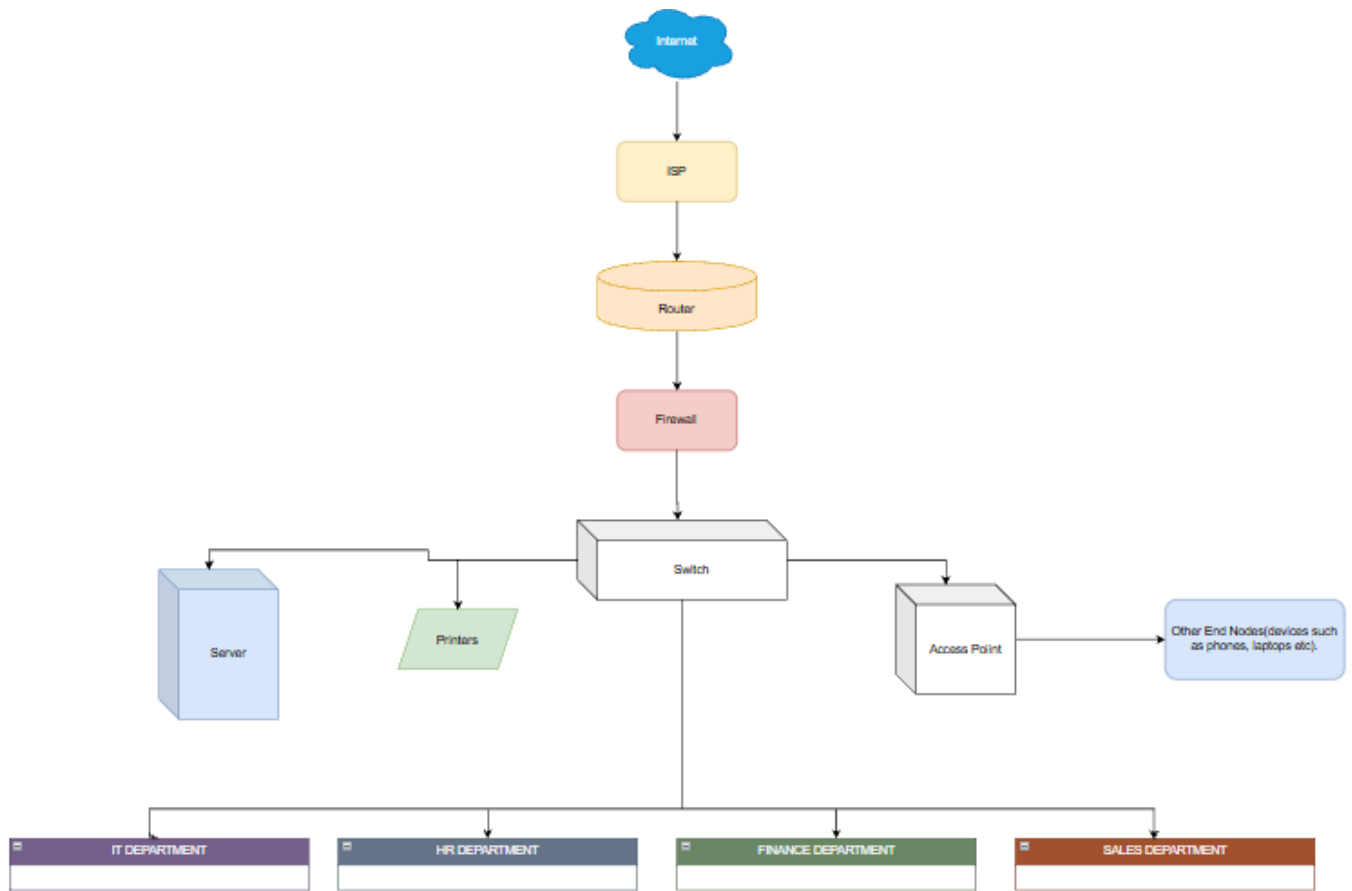
Software	Version	License	Purpose
Windows 11 Pro	Latest	Commercial	Primary OS for Employess
Ubuntu Server	24.04 LTS	Open Source	Server OS for Hosting internal Services
MS Office	Latest	Subscription	Document, CSV and Presentations
VSCODE	Latest	Free	Source Code editing
Git	Latest	Open Source	Version Control and Source code editing
Github	Latest	Free	Interface for managing repo and git
VirtualBox	Latest	Open Source	Virtual machine creation and testing environment
Google Chrom	Latest	Free	Web browsing
Microsoft Defender	Latest	Comes with windows	Baseline Antivirus or EDR
Anydesk	Latest	Commercial	Remote Desktop Access
7-zip	Latest	Commercial	File Compression and archive management

Enterprise Network Inventory

Item	Quantity	Specification	Purpose
ISP Modem	1	Fiber Optic Modem	Provides connectivity from the ISP
Router	1	Gigabit Business Router	Routes Traffic and Enforces Policies
Firewall	1	Hardware Firewall Appliance	Protects the network from unauthorized access
Switch	1	24-port Gigabit Switch	Connects network Devices
Access-point	3	Dual Band wireless access point	Provides wireless connectivity
Patch Panel	1	24-port cat6	Organizes and manages network cabling
Cat6 cables	30	cat6 ethernet cables	connects computers, printers and other devices
RJ45 Connectors	50	Cat6 Compatible	Cable termination and network installation

Enterprise Network Diagram

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System Admin Roles

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Helpdesk Technician

Responsibilities

Providing technical support to end users, troubleshooting hardware and software issues, installing and configuring computer systems, managing user accounts and passwords, and escalating complex technical problems when necessary.

Skills

- Troubleshooting and problem-solving
- Customer service and communication
- Basic networking knowledge
- Operating system administration
- Time management

Common Tools

- AnyDesk
- Microsoft Remote Desktop
- Ticketing Systems (e.g., Freshdesk, ServiceNow)
- Microsoft Defender
- Windows Tools and Utilities

Certifications

- CompTIA A+
- Google IT Support Professional Certificate
- ITIL Foundation

Network Administrator

Responsibilities

Managing and maintaining network infrastructure, configuring routers and switches, monitoring network performance, implementing network security measures, and resolving connectivity issues across the organization.

Skills

- TCP/IP Networking
- Routing and Switching
- Network Security
- Network Monitoring
- Troubleshooting and Analysis

Common Tools

- Cisco Packet Tracer
- Wireshark
- PRTG Network Monitor
- Cisco IOS
- SolarWinds

Certifications

- Cisco Certified Network Associate (CCNA)
- CompTIA Network+

- Cisco Certified Network Professional (CCNP)

Linux System Administrator

Responsibilities

Installing and maintaining Linux servers, managing users and permissions, monitoring system performance, performing backups and updates, and ensuring the security and reliability of server environments.

Skills

- Linux Command Line
- Shell Scripting
- Server Administration
- System Monitoring
- Security Management

Common Tools

- SSH
- Bash
- Ansible
- Git
- Ubuntu Server

Certifications

- Red Hat Certified System Administrator (RHCSA)
- CompTIA Linux+
- Linux Professional Institute Certification (LPIC-1)

Cloud Administrator

Responsibilities

Managing cloud infrastructure, deploying and monitoring cloud resources, maintaining cloud security, implementing backup and disaster recovery solutions, and optimizing cloud services for performance and cost efficiency.

Skills

- Cloud Computing
- Virtualization
- Cloud Security
- Automation and Scripting
- Resource Management

Common Tools

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)
- Terraform
- Cloud Monitoring Tools

Certifications

- AWS Certified Solutions Architect – Associate
- Microsoft Certified: Azure Administrator Associate
- Google Associate Cloud Engineer

How These Professionals Work Together

At DRWN Technologies, the Helpdesk Technician serves as the first point of contact for employees who encounter technical issues, ensuring that problems are resolved quickly to minimize disruptions. The Network Administrator is responsible for maintaining a stable and secure network infrastructure that supports communication and connectivity across all departments. The Linux System Administrator manages the company's servers, user accounts, and critical services to ensure reliable system performance and data availability. Meanwhile, the Cloud Administrator oversees cloud resources, backups, and online services that support the organization's operations. By working closely together, these professionals ensure that the company's IT infrastructure remains secure, efficient, and capable of supporting both current business needs and future growth.

Recommendations

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Internet Provider

For DRWN Technologies, a business fiber internet plan with a minimum speed of **300 Mbps** is recommended. Since the company relies on cloud services, software development platforms, online collaboration tools, and remote communication, a stable and high-speed internet connection is essential. A business-grade plan also provides better reliability, dedicated technical support, and room for future expansion.

Server Specifications

DRWN Technologies should deploy a dedicated enterprise server equipped with an **Intel Xeon or AMD EPYC processor**, **32 GB of RAM**, and **dual 1 TB SSDs configured in RAID 1** for improved performance and data redundancy. The server should run **Ubuntu Server 24.04 LTS** to provide a stable and secure environment for file sharing, user account management, backups, and other internal services. This configuration offers sufficient resources for the company's current workforce while allowing for future growth as additional employees and services are added.

Backup Strategy

A **3-2-1 backup strategy** is recommended to ensure data protection and business continuity. The company should maintain three copies of important data, store backups on two different storage media, and keep one backup copy off-site or in the cloud. Daily incremental backups and weekly full backups should be performed to minimize data loss in the event of hardware failure, accidental deletion, or cybersecurity incidents.

Security Recommendations

To strengthen cybersecurity, DRWN Technologies should implement a dedicated firewall, keep all systems updated with the latest security patches, and enforce role-based access control. Multi-factor authentication should be enabled for critical accounts, and employees should receive regular cybersecurity awareness training. These measures will help reduce security risks and protect company resources from unauthorized access.

Antivirus

Microsoft Defender for Business is recommended as the company's primary antivirus solution. It provides real-time protection against malware, ransomware, phishing attacks, and other cybersecurity threats while integrating seamlessly with Windows 11 Pro devices used throughout the organization.

Expansion Plan

As DRWN Technologies grows, its infrastructure should be designed to scale efficiently. Additional workstations, access points, network switches, and

storage solutions can be added as employee numbers increase. The company should also consider expanding its cloud services and upgrading server resources when demand rises. Planning for growth early will allow the organization to accommodate future business requirements without major disruptions or costly infrastructure redesigns.

Reflection

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Working on this project taught me a lot about IT infrastructure planning and the responsibilities of a System Administrator. Before doing this activity, I thought setting up an IT environment was mostly about installing computers and connecting them to the internet. However, I learned that there is a lot of planning involved before any equipment is purchased or deployed. I learned how to create hardware, software, and network inventories, design a network topology, and recommend solutions that fit the needs of a company.

The most challenging part of this project was creating the network diagram. I had to carefully think about how each device would connect to the network and how the different departments would communicate with each other. It took several revisions before I was satisfied with the layout and made sure that all required components were included. Although it was difficult at first, it helped me better understand how enterprise networks are structured.

Planning is important before deployment because it helps organizations avoid unnecessary costs, reduce technical problems, and improve security. Without proper planning, a company may purchase the wrong equipment, experience network issues, or face difficulties when expanding in the future. Having a clear infrastructure plan makes it easier to identify requirements, allocate resources, and ensure that all systems work together efficiently.

This project will help me become a better System Administrator because it gave me practical experience in planning and documenting an IT infrastructure. It improved my understanding of how hardware, software, networking, and security all work together to support business operations. It also helped me develop my problem-solving and decision-making skills when selecting equipment and designing solutions. Overall, this project gave me a better appreciation of the work involved in managing and planning an organization's technology environment, and it has increased my confidence in pursuing a career in system administration.

